Dina Saad Ahmed Mohammed

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Professional Profile

A highly analytical and growth-oriented Academic Educator and Electronics Engineer with a unique combination of academic research depth and full-cycle hardware prototyping expertise. Masterful in translating complex mathematical algorithms and schematic concepts into manufacture-ready physical designs, with specialized competencies spanning advanced power electronics, multi-layer industrial PCB technologies, full-custom analog IC layout, and digital signal processing frameworks. An experienced technical instructor with a proven track record of mentoring large student cohorts through hardware simulation, troubleshooting, and real-world system realization. Possesses high emotional intelligence and a dedicated growth mindset, excelling in collaborative, multi-disciplinary, and highly technical environments.

Technical Skill Set

Hardware Engineering & Industrial PCB Design

- **PCB Design Tools:** Altium Designer, EasyEDA.
- **Board Technologies:** Multi-layer Rigid, Flexible (Flex), and Rigid-Flex PCB architectures.
- **High-Density Interconnect (HDI):** Specialized routing using Through-hole, Blind, Buried, and Micro-via topologies.

- **Design Practices:** Design for Manufacturing (DFM) and Design for Assembly (DFA) optimization, component parameter extraction from datasheets, component sizing, high-speed routing, and signal/power integrity.
- **Power Electronics & Protection:** Isolated topologies (Flyback), non-isolated modeling (Buck), closed-loop control (PID tuning, Type 2/3 compensators), Voltage/Current Mode Control, gate drivers, snubber design, EMC filter design, and hardware protection (overcurrent, overvoltage, ESD, thermal,polarity).
- **Circuit Simulation:** LTSpice, MATLAB/Simulink, Proteus, Multisim, Circuit Wizard.

Academic Instruction & Digital Signal Processing (DSP)

- **Signal & System Theory:** Continuous-to-Discrete sampling, quantization boundaries (ADC/DAC constraints), anti-aliasing/anti-image filtering, LTI system characterization, causality, and BIBO stability verification.
- **Transform Domains & Filtering:** Discrete Fourier Transform (DFT), Fast Fourier Transform (FFT via Decimation-in-Time/Frequency), Z-Transforms, FIR filter synthesis (Window, Frequency Sampling, Optimal), and IIR filter design (Bilinear Transformation, Impulse Invariant, Prototype Mapping).
- **Applied DSP Paradigms:** Spectral estimation, signal enhancement (60-Hz hum elimination, speech bandpass filtering, ECG notch filtering), and dual-tone multifrequency (DTMF) detection using the Goertzel/Modified Goertzel algorithms.

Analog & Mixed-Signal IC Design

- **Design Flows:** Full-Custom Layout, Transistor Sizing, Schematic Capture.
- **Physical Verification:** Design Rule Check (DRC), Layout Versus Schematic (LVS), Parasitic Extraction (PEX).
- **EDA Software Tools:** Cadence Virtuoso, L-Edit.

Digital Systems, RF, & Core Software

- **Digital Logic Development:** VHDL, Verilog, System Verilog, FPGA deployment workflows.
- **Synthesis & Emulation Tools:** Xilinx ISE, QuestaSim.
- **RF & Antenna Simulation:** Ansys HFSS, CST Studio Suite, Microstrip Patch Antenna engineering, Impedance Matching.
- **Programming Languages:** C, C++, Python, MATLAB, Object-Oriented Programming (OOP) paradigms.
- **Mechanical Modeling:** SolidWorks (electromechanical enclosure design and physical system integration).

Professional & Academic Work Experience

Electronics Instructor | Medical Electronics Company, Asyut

May 2024 – Present

- Managed the end-to-end laboratory and theoretical training for over 100 engineering students and graduates, introducing them to practical circuit layouts and industrial assembly standards.
- Designed and delivered a comprehensive curriculum covering the operational mechanics of hardware systems, focusing on battery charging architectures and linear power supplies.

- Mentored students through software-level circuit simulation inside Proteus and LTSpice before supervising their physical, bench-top board prototyping.

University Instructor (Freelance)

- Delivered a high-level undergraduate course in Digital Signal Processing (DSP), successfully bridging abstract mathematical proofs with practical **MATLAB** algorithmic modeling.
- Instructed students on the deep theoretical and simulation foundations of sampling, quantization noise, discrete convolution, difference equations, and system representation using impulse and step responses.
- Guided sessions on spectral analysis (DFT/FFT methods), Z-transform applications, and structural digital filter realizations (Direct-Form I & II, Cascade, Parallel, Transversal, and Linear Phase forms).

Electronics Demonstrator | Asyut Technological University, Asyut

Oct 2024 – Jan 2025

- Co-instructed large-scale laboratory modules for a cohort of over 300 technical undergraduate students in coordination with senior university faculty.
- Taught, demonstrated, and troubleshot discrete electronic circuit topologies, including operational audio amplifiers and automated sensor-driven switching systems.
- Leveraged digital interactive platforms (Ziteboard, PowerPoint) to communicate abstract hardware behavior, significantly boosting student performance and analytical retention.

Education & Advanced Engineering Training

National Telecommunication Institute (NTI)

Professional Internship: Electronics & PCB Design for Industry | 2026

- **NTI Capstone Project (USB-C Phone Charger):** Engineered an isolated AC-DC Type-C mobile phone charger using a **Flyback Converter topology** from scratch. Performed thorough transformer parameter sizing, primary-to-secondary isolation layout routing, EMC filter design and snubber loop calculations to achieve high efficiency within a strict physical form-factor constraint.
- **Control Systems & Converter Modeling:** Built non-ideal small/large-signal models of Buck Converters to evaluate steady-state regulation. Designed closed-loop control systems utilizing **PID controllers, Type 2, and Type 3 compensators** inside MATLAB/Simulink and LTSpice. Handled diverse control paradigms including VMC, CMC, CCM, and DCM.
- **Industrial Manufacturing (DFM/DFA):** Developed complex PCB physical layouts across multi-layer **Rigid, Flex, and Rigid-Flex** substrates. Managed HDI via stackups (blind, buried, micro-vias) and generated fabrication deliverables: Gerbers (X2), NC Drill files, IPC-compliant footprint libraries, and detailed Bills of Materials (BOM).
- **Hardware Protection Design:** Evaluated discrete semiconductors and engineered gate drive circuits for power MOSFETs. Implemented protection circuits for overcurrent (fuses, poly-

switches, NTCs), overvoltage (Zener, TVS), polarity reversals, ESD, and thermal runaway.

Master’s Degree in Wireless Communication

Assiut University, Asyut, Egypt | Feb 2025 – Present

- **Research Specialization:** Active Reconfigurable Intelligent Surfaces (ARIS)-Assisted Integrated Sensing and Communication (ISAC) Systems for 6G Networks.
- Conducting deep mathematical optimization and signal model simulations for next-generation telecommunication infrastructures.

Bachelor’s Degree in Electronics and Communication Engineering (Honors)

Assiut University, Asyut, Egypt | Graduation Date: June 2023

- **Overall Cumulative Grade:** Very Good (83.2%)
- **Graduation Project:** Reinforcement Learning-Based Optimization for UAV-Borne RIS Wireless Communication Systems.
- **Project Grade:** Excellent (93.33%).

Key Engineering & Research Projects

Full-Custom Analog IC Design Flows

- **CMOS Logic NOT Gate (Cadence Virtuoso):** Designed an inverter schematic down to the transistor level; performed symmetrical threshold scaling, executed DC/Transient analyses in ADE, and ran Parasitic Extraction (PEX) to ensure timing verification.

- **Custom CMOS Layout Optimization (L-Edit & SPICE):** Applied the Euler Path layout technique (ABECD) to implement continuous diffusion strips, minimizing silicon area. Ran physical verifications including Design Rule Checks (DRC) and Layout Versus Schematic (LVS) to eliminate errors, achieving 100% netlist matching and characterizing precise post-layout rise/fall delays.

Digital Logic & Hardware Co-Design

- **VHDL Processor Modules (Xilinx ISE):** Designed and verified hardware logic blocks, including an 8-bit Arithmetic Logic Unit (ALU) capable of executing 16 hardware operations with carry flags, address routing decoders (74LS138), and robust Finite State Machines (FSM) for controllers.
- **Remotely Operated Underwater Vehicle (ROV):** Engineered an underwater robotic vehicle balancing mechanical layout constraints, power distribution, and automated control. Programmed real-time embedded computer vision algorithms using **OpenCV** for automated pathing and underwater target tracking.

RF Modeling & Intelligent Infrastructure

- **Inset-Fed Microstrip Patch Antenna:** Modeled and simulated a rectangular microstrip patch antenna optimized for a 1.65 GHz resonant frequency inside Ansys HFSS and CST Studio Suite; resolved return-loss mismatches using an inset feed.
- **Smart Fleet IoT Ecosystem Business Case:** Developed an end-to-end smart public transportation infrastructure plan for Egypt, defining technical telemetry hardware constraints, data-routing protocols, and formal project charters.

Volunteering Work & Student Leadership

- **Electric Academy Member** | Asyut Robotics Student Team (Jun 2019 – Sep 2022)
- **Team Organizer** | Ethad Qawafel Handasa Charity (Apr 2019 – Present)
- **Organizer** | Asyut Blood Donors Union (ABDU) (Apr 2023 – Present)